Ka Ming FUNG, Ph.D.

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EDUCATION BACKGROUND

The Chinese University of Hong Kong (CUHK)

Doctor of Philosophy in Earth and Atmospheric Sciences — *Aug 2015 – Aug 2019*

- *Thesis Title*: Modeling and Assessing the Impacts of Sustainable Agricultural Practices on Air Quality and Climate

Master of Science in Mathematics (*Final GPA*: 3.688/4) — *Sep 2011 – Jun 2012*

Hong Kong University of Science and Technology (HKUST)

Bachelor of Engineering in Mechanical Engineering (Second Class Honor, Division I) — *Sep 2006 – Jun 2009*

RESEARCH EXPERIENCE

Postdoctoral Associate, Department of Civil and Environmental Engineering, Massachusetts Institute of Technology (MIT) — *Sep 2019 – Present*

- expanded the chemistry mechanisms in the Community Atmosphere Model with chemistry (CAM-chem) to better represent dimethyl sulfide (DMS) oxidation
- evaluated the impact of DMS chemistry on aerosol formation and climate forcing

Visiting Researcher, Leverhulme Centre for Climate Change Mitigation, Department of Animal and Plant Sciences, The University of Sheffield — *Mar – May 2019*

- developed a new scheme to quantify the canopy capture of ammonia emission from vegetated land and croplands in the Community Land Model (CLM)
- validated simulated ammonia against various satellite & in-situ observations and emission inventories

Visiting Scholar, Department of Environmental Sciences, Emory University — *Oct 2017 – Mar 2018*

- implemented into CLM novel parameterizations of belowground crop-crop competition and soil ammonia volatilization
- assisted in soil sample collection and heavy metal measurement for a feasibility assessment of urban farming in Atlanta, Georgia, USA

Junior Research Assistant, Earth System Science Programme, CUHK — *Apr – Jul 2015*

- studied global impacts of agricultural activities using a biogeochemical model DeNitrification-DeComposition (DNDC), a 3-D global chemical transport model (GEOS-Chem), and the Community Earth System Model (CESM)
- conducted data analysis on agricultural nitrogen emissions, crop production, and soil biogeochemical processes using R

Junior Research Assistant, Department of Imaging and Interventional Radiology, CUHK — *Mar 2013 – Mar 2015*

- published a paper in *Scientific Reports* (*DOI*: 10.1038/srep14140)
- prepared proposals, gave presentations to the selection committees, and obtained research grants totaled HKD3.2M from the Innovation and Technology Fund (*Project Ref.*: ITS/149/14FP and ITS/293/14FP)
- built MATLAB tools for mathematical and statistical analyses on four-dimensional medical images
- coordinated a 40-subject study involving MRI brain scanning and cognitive tests

PUBLICATIONS

Journal articles:

1. K.M. Fung, A. P. K. Tai & M. Val Martin (*preprint*) **Modeling the interinfluence of fertilizer-induced NH_3 emission, nitrogen deposition, and aerosol radiative effects using modified CESM2**, *Biogeosciences Discussions* (*DOI:* 10.5194/bg-2021-63)

2. K. M. Fung, A. P. K. Tai, T. Yong, X. Liu & H.-M. Lam (2019) **Co-benefits of intercropping as a sustainable farming method for safeguarding both food security and air quality**, *Environmental Research Letters* (*DOI*: 10.1088/1748-9326/aafc8b)

3. D. Wang, F. Zhu, K. M. Fung, W. Zhu, Y. Luo, W. C. W. Chu, V. C. T. Mok, J. Wu, L. Shi, A. T. Ahuja & Y. Mao (2015) **Predicting Cerebral Hyperperfusion Syndrome Following Superficial Temporal Artery to Middle Cerebral Artery Bypass based on Intraoperative Perfusion-Weighted Magnetic Resonance Imaging**, *Scientific Reports* (*DOI*: 10.1038/srep14140)

PRESENTATIONS

Invited Talks:

1. **Modelling and Assessing the Impacts of Intercropping, as a Sustainable Farming Practice, on Food Security, Air Quality, and Public Health**, *Centre for Atmospheric Science Seminars*, Department of Chemistry, University of Cambridge, Cambridge, UK (May 2019)

Conference Talks:

1. **Exploring natural aerosol formation from DMS oxidation and implications for aerosol forcing**, *CESM Atmosphere, Whole Atmosphere, and Chemistry-Climate Working Group Meeting 2021*, online (Feb 2021)

2. **Impacts of Ammonia-Aerosol-Climate Feedbacks on Food Security and Air Quality**, *CESM Land Model Working Group Meeting 2020*, Boulder, Colorado, USA (Mar 2020)

3. **Modeling ammonia-aerosol-climate feedback mechanisms using an earth system model: Implications for future food security and air quality**, *American Geophysical Union (AGU) Fall Meeting 2019*, San Francisco, California, USA (Dec 2019)

4. **Modeling and Assessing the Impacts of Sustainable Farming Practices on Food Security, Air Quality, and Public Health**, *The Chemistry Climate Model Initiative (CCMI) Science Workshop*, Hong Kong (Aug 2019)

5. **Improving the terrestrial N cycle modeling for a better estimation of agricultural ammonia emission under sustainable farming alternatives**, *AGU Fall Meeting 2018*, Washington DC, USA (Dec 2018)

6. **Large-scale adoption of intercropping for securing global food supply and air quality – a model study using CLM 4.5**, *CESM Land Model Working Group Meeting 2018*, Boulder, Colorado, USA (Feb 2018)

7. **Modeling large-scale adoption of intercropping as a sustainable agricultural practice for food security and air pollution mitigation around the globe**, *AGU Fall Meeting 2017*, New Orleans, Louisiana, USA (Dec 2017)

8. **Modeling and assessing effectiveness of intercropping as a sustainable agricultural practice for food security and air pollution mitigation**, *5th Integrated Land Ecosystem-Atmosphere Processes Study (iLEAPS) Science Conference*, University of Oxford, Oxford, UK (Sep 2017)

Conference Posters:

1. **Exploring natural aerosol formation from DMS oxidation and implications for aerosol forcing**, *AGU Fall Meeting 2020*, online (Dec 2020)

2. **Co-benefits of Intercropping, as a Sustainable Farming Practice, for Safeguarding Food Supply and Air Quality**, *Gordon Research Conference – Urbanization, Water and Food Security*, HKUST, Hong Kong (Jul 2019)

3. **Estimation of agricultural ammonia emission under sustainable farming practices by improving terrestrial nitrogen cycle modeling**, *7th Annual Composition-Climate Interaction Meeting*, Exeter, UK (Mar 2019)

4. **Co-benefits of maize-soybean intercropping for securing air quality and global food supply**, *Global Young Scientists Summit 2019*, Singapore (Jan 2019)

5. **Potential co-benefits of intercropping as a sustainable agricultural practice for both air pollution mitigation and global food security**, *Joint 14th iCACGP Symposium and 15th IGAC Science Conference*, Takamatsu, Japan (Sep 2018)
6. **Evaluating effectiveness of maize-soybean intercropping in securing food production and air quality in China using DNDC and GEOS-Chem**, *The 8th International GEOS-Chem Meeting*, Harvard University, Massachusetts, USA (May 2017)
7. **Effectiveness of intercropping with soybean as a sustainable farming practice to maintain food production and reduce air pollution in China**, *AGU Fall Meeting 2016*, San Francisco, California, USA (Dec 2016)

TEACHING EXPERIENCE

Guest Lecturer, Graduate Division of Earth and Atmospheric Sciences, CUHK
- EASC5001 Research Frontiers in Earth & Atmospheric Sciences I (a special lecture on nitrogen cycle modeling) *Fall 2018*

Mentor, Earth System Science Programme, CUHK
- ESSC4820 Senior Project II *Spring 2017*
- ESSC4810 Senior Project I *Fall 2016*

Teaching Assistant, Earth System Science Programme, CUHK
- ESSC4520 Numerical Methods and Modeling for Earth and Atmospheric Sciences *Spring 2016*
- ESSC2020 Climate System Dynamics *Fall 2015*

AWARDS & HONORS

Trusted Reviewer, IOP Publishing Limited *2020*

Travel Support for *CESM Land Model Working Group Meeting 2020*, National Centre for Atmospheric Research (NCAR) *2020*

Li Po Chun Charitable Trust Fund Postgraduate Scholarship *2019*

Travel Support, *7th Annual Composition-Climate Interaction Meeting*, National Centre for Atmospheric Science (NCAS), University of Leeds *2019*

Global Scholarship Programme for Research Excellence, Office of Academic Links, CUHK *2019*

Selected Attendee, *Global Young Scientists Summit 2019*, the Singapore Government *2019*

Travel Grant and Selected Attendee of the Early Career Short Course, *Joint 14th iCACGP Symposium & 15th IGAC Science Conference*, IGAC Project *2018*

Pre-doctoral Fellowship, Institute for Quantitative Theory and Methods, Emory University *2017*

Reaching Out Award, HKSAR Government *2017*

Winner of The Young Scholar Competition, *WUN Symposium cum Research Summit – Impacts of Grain Legume Research and Development in Developing Countries*, CUHK *2017*

Service Award, Earth System Science Programme, CUHK *2015/16 & 2016/17*

OTHER WORK EXPERIENCE

Mathematics Teacher, NTK Academic Group *Oct 2012 – Mar 2013*
- specialized in mathematics of IB, GCE, I/GCSE curricula at various levels
- wrote study guides & exercise books

Management Trainee, The Jardine Engineering Corporation, Limited *Jul 2009 – May 2011*
- managed a project to build the largest cremation system in Hong Kong
- self-initiated to automize the workflow for maintaining a database of government contracts using VBA for Excel

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INTERNSHIP

Co-op Student Trainee, The Jardine Engineering Corporation, Limited *Feb – Jun 2009*

- earned a return offer to join the company at the end of this internship
- participated in tender bidding and helped to win a contract worth HKD118M (*Ref.*: SST326, Architectural Services Department, HKSAR Government)

PROFESSIONAL ACTIVITIES

Host & Organizer, *Diversity and Inclusion Seminar Series*, Department of Civil and Environmental Engineering, MIT *2020*

Conference/Workshop Organizing Committee

- *CCMI Summer School & Science Workshop*, IGAC *2019*
- *Early Career Scientist Workshop, 5th iLEAPS Science Conference* *2017*
- *Workshop on Modeling Land-Atmosphere Interactions: Fire, Agricultural and Canopy Processes*, CUHK *2017*

Guest Editor

- co-edited a special issue in *Frontiers in Climate*: Agriculture and Food Supply Challenges in a Changing Climate *2021*

Journal Reviewer

- reviewed 11 papers in *Environmental Research Communications*, *Environmental Research Letters*, *Global and Planetary Change* & *Journal of Geophysical Research: Biogeosciences* *Since 2019*

Regional Head (SE Asia & Pacific), Early Career Scientist Regional Network, iLEAPS *2018 – Present*

COMMUNITY SERVICE

Postdoctoral Liaison, Department of Civil and Environmental Engineering, MIT *2020 – Present*

Resident Tutor & Honorary Resident Tutor, Daisy Li Hall, New Asia College, CUHK *2013 – 2019*

Councilor, HKUST Students' Union (SU) Council *2008 – 2009*

Student Ambassador, School of Engineering, HKUST *2008 – 2009*

Publication Secretary, Glacier, HOUSE III Students' Association, HKUSTSU *2007*

LANGUAGES

Chinese	fluent written, native Cantonese and conversational Mandarin
English	fluent spoken and written, attended IELTS in 2008 with an overall score of 7.0

COMPUTER SKILLS

Scientific Modeling	CESM (CAM-chem & CLM), DNDC, GEOS-Chem, WRF
Programming	C++, Fortran, MATLAB, Python, R, VBA for Excel
Graphical Drawing	Adobe Illustrator, Photoshop
Technical Drawing	Autodesk 3ds Max, AutoCAD, SolidWorks
Video Editing	Adobe Premiere
Website Design	Hugo, Jekyll